

Topology General Exam January 2014

Solve the following problems on your own paper. Be sure your solutions are legible and clearly organized. All work should be your own; no outside sources are permitted. You may use without proof standard results from first-semester differential and algebraic topology; where appropriate you should cite theorems by name.

Problem 1

Suppose X is a connected, locally path-connected space whose fundamental group is finite. Prove that every continuous map from X to the torus T is null-homotopic.

Problem 2

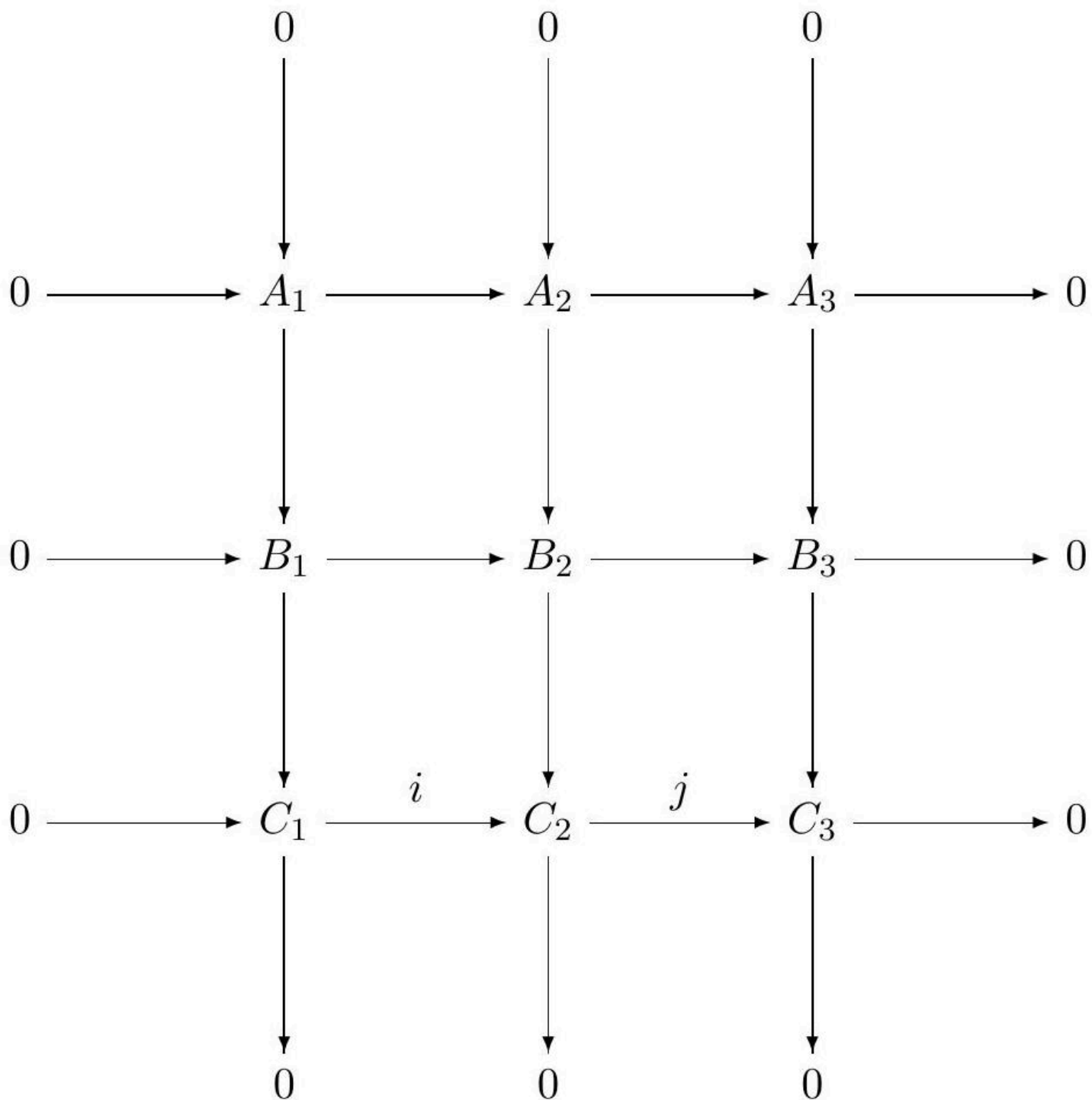
Let T_1 and T_2 be two copies of the torus $S^1 \times S^1$, and let $f, g : S^1 \rightarrow S^1$ be two maps of degrees 2 and 6, respectively. Let $x_0 \in S^1$ be a fixed base point. Find the fundamental group of the space

$$X = T_1 \cup_F S^1 \times [0, 1] \cup_G T_2,$$

where $F : S^1 \times \{0\} \rightarrow T_1$ is given by $F(x, 0) = (f(x), x_0)$ and $G : S^1 \times \{1\} \rightarrow T_2$ is given by $G(x, 1) = (g(x), x_0)$.

Problem 3

You are given the following commutative diagram of abelian groups:



The rows are $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$, $0 \rightarrow B_1 \rightarrow B_2 \rightarrow B_3 \rightarrow 0$, and $0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow 0$. Each column is $0 \rightarrow A_r \rightarrow B_r \rightarrow C_r \rightarrow 0$, directed downward, for $r = 1, 2, 3$. In the bottom row, i maps C_1 to C_2 and j maps C_2 to C_3 . Every square commutes.

Assume that all columns are exact, and that the first two rows are exact. Prove that the third row is exact at C_2 , i.e., that $\text{im}(i) = \ker(j)$. (Note: in this situation the entire third row is exact, but you need not prove this.)

Problem 4

Consider a space X that is the union of two open subsets U and V such that:

- V is contractible
- both U and $U \cap V$ have the homology of a circle.

- a) What are all the possibilities for the homology $H_*(X; \mathbb{Z})$?
- b) Describe explicit spaces realizing all these possibilities.

Problem 5

Let M be a smooth compact manifold without boundary. Show that there is no submersion (i.e., smooth map whose differential is everywhere surjective) $F : M \rightarrow \mathbb{R}^k$ for any $k > 0$.

Problem 6

Show that for any $n \geq 0$ the manifold $M = S^n \times \mathbb{R}$ is parallelizable (that is, its tangent bundle is trivial).

Problem 7

Let M be a smooth, closed (compact without boundary) n -dimensional submanifold of $\mathbb{R}^{n+1}$, with $0 \notin M$. Prove that there exists a line through 0 in $\mathbb{R}^{n+1}$ which intersects M in finitely many points (or is disjoint from M).

Problem 8

Let M and N be the subsets of $\mathbb{R}^3$ defined by

$$M = \{x^2 + y^2 + z^2 = 1\} \quad N = \{x^2 - y^2 + z^2 = c\}$$

for a real number c . Justify your responses to the following:

- a) Determine all values of c for which M and N are submanifolds of $\mathbb{R}^3$, and the intersection $M \cap N$ is transverse.
- b) Determine all values of c for which $M \cap N$ is a submanifold of $\mathbb{R}^3$.

Problem 9

Suppose $f : S^n \rightarrow S^n, n \geq 2$, is a smooth map whose differential is injective at each point. Prove that f is a diffeomorphism.